

Lecture 10

Errors, Failures, and Risks

Understanding the risks and reasons
for computer failures.

Interface Design Perspective

- To the user “the interface is the system”
 - Understanding of the system operation is based on its interface
 - If this conceptual model is wrong, errors result
- Human attention is incomplete – will not notice every detail, every notification
- Good design will help prevent human error and help users recover from errors

Interface Design Perspective

- Beware of “response chaining”
 - Example: Nurse accidentally turned off alarm on monitoring for critically ill patient

What goes wrong?

- Computer systems fail because:
 - The job they are doing is inherently difficult
 - The job is sometimes done poorly
- Compounding the reliability issue
 - Developers and users are overconfident in the system
 - Re-used system software may not work correctly in different environments
- Intellectual overload
 - Large bodies of code are more difficult to understand than smaller ones

What can go wrong?

- Q: How can we increase reliability & safety?
 - As developers?
 - As system managers?
 - As users?

Increasing reliability and safety

- Be professional and responsible:
 - Follow good software-engineering practices
 - What do you believe constitutes “good”?
- Construct well-designed user interfaces + take human factors into account
- Incorporate self-checking where appropriate
- Follow good testing principles and techniques

Increasing reliability and safety

- Law and regulation
 - Civil and criminal penalties
 - to recover from loss due to faulty or unsafe systems
 - Warranties
 - to guarantee a certain level of quality
 - Mandatory licensing of software developers
 - to ensure proper training, competency, and continuing education

Computer Models

- Points to consider
 - Models are simplifications of either physical or intangible systems
 - Those who design and develop models must be honest and accurate with results
 - Computer professionals and the general public must be able to evaluate the claims of the developers

Computer Models

- Evaluating models
 - Why models might not be accurate:
 - Developers have incomplete knowledge of the system being modeled
 - Data might be incomplete or inaccurate
 - Power of the computer might be inadequate
 - Political and economic motivation to distort results
 - Two contrasting examples:
 - Car-crash models
 - Climate models

Computer models

- Car-crash models
 - How well do the modelers understand the system or material being studied (or both)?
 - How accurate and complete are the data?
 - Do the results or predictions correspond with results in the real world?
- Climate models
 - (Same questions)

Thank You